JavaScript Fundamentals — Day 1

Detailed beginner notes based on what you have learned so far

1. What is JavaScript?

JavaScript (JS) is a programming language commonly used to make web pages interactive and dynamic.

HTML → structure of the webpage
CSS → appearance and styling
JavaScript → behaviour, interaction and logic

Examples of JavaScript behaviour: responding to button clicks, calculating values, checking form input, changing webpage content, showing messages, and handling user actions.

2. Your First JavaScript Output

The most basic output statement you learned is **console.log()**.

```
console.log("Hello");
```

What it means: Ask JavaScript to display the given value in the browser's developer console.

```
console.log(25);
```

The quotation marks matter: `"Hello"` is text (a String), while `25` is a Number.

3. Variables

A **variable** is a named place used to store a value. You can think of it like a labelled box.

```
let name = "Namratha";
```

let → declares the variable
name → variable name
= → assignment operator; puts the value into the variable
"Namratha" → value stored in the variable

After creating a variable, you can use its name instead of repeatedly writing the value.

```
let age = 21;
console.log(age);
```

4. let, const and var

Keyword	Main idea	Example	Can be reassigned?
let	Use for a variable whose value may change.	let age = 21;	Yes
const	Use when the variable should not be reassigned.	const pi = 3.14;	No
var	Older way of declaring variables.	var city = "Bangalore";	Yes (but avoid for modern code)

Easy memory: `let` = let it change; `const` = constant/no reassignment. `var` is the older declaration style.

Important: `const` means the variable cannot be reassigned. It does not mean every object or array stored in a const variable is completely immutable; that is a later topic.

5. Data Types

A **data type** tells JavaScript what kind of value something is. The main types you learned today are:

Type	Meaning	Example
String	Text	"Hello"
Number	Numeric values, including decimals	25 / 3.14
Boolean	Only true or false	true
Undefined	A variable exists but has not been assigned a value	let x;
Null	Intentionally represents no value	let user = null;

String

A String is text. Strings are commonly written inside quotes, such as `"Hello"` or `"123"`.

> **Important:** `"123"` is a String, while `123` is a Number.

Number

Numbers can be used for calculations. Examples: 10, 21, 99.5, -4.

Boolean

A Boolean has only two possible values: `true` or `false`. It is useful for yes/no or on/off conditions.

Undefined vs Null

`let name;` → the variable has no assigned value, so it is **undefined**.
`let selectedUser = null;` → you intentionally set it to have no value.

6. Operators

Operators are symbols that tell JavaScript to perform an operation.

Arithmetic operators

Operator	Meaning	Example	Result
+	Addition	10 + 3	13
-	Subtraction	10 - 3	7
*	Multiplication	10 * 3	30
/	Division	10 / 3	3.333...
%	Remainder	10 % 3	1

Remember %: It gives the remainder after division. For example, 10 divided by 3 leaves 1.

Comparison operators

> greater than
< less than
>= greater than or equal to

<= less than or equal to
== loose equality
=== strict equality
!= loose inequality
!== strict inequality

> **Important:** You will commonly prefer === and !== because they compare without the type conversion associated with loose equality.

Logical operators

&& → AND: both conditions need to be true.
|| → OR: at least one condition needs to be true.
! → NOT: flips true to false and false to true.

7. Type Conversion

Type conversion means changing a value from one data type to another. This is especially important when working with user input.

```
Number("25")
```

Converts the String `"25"` into the Number `25`.

```
String(25)
```

Converts the Number `25` into the String `"25"`.

```
Boolean(1) → true
Boolean(0) → false
```

8. User Input with prompt()

`prompt()` opens a browser dialog that asks the user for input and returns what the user entered.

```
let name = prompt("Enter your name:");
```

Very important: the value returned by `prompt()` is a String.

Therefore, when you ask for a number, convert the result if you want to perform reliable numeric calculations:

```
Number(prompt("Enter your age:"))
```

9. String Concatenation and Type Coercion

You discovered an important JavaScript behaviour while testing your calculator.

```
"10" + 5 → "105"
```

The + operator can concatenate (join) text. JavaScript treats the number as text for this operation.

By contrast, mathematical operators generally convert a numeric string to a number:

```
"5" - 2 → 3
"5" * 2 → 10
"5" / 2 → 2.5
```

Best beginner habit: when you expect a number from `prompt()`, explicitly convert it using `Number()`.

10. Your Calculator Practice

You built the basic HTML structure for a calculator and started connecting it to JavaScript. Your calculator uses HTML for the buttons/display and JavaScript for the calculations.

The JavaScript-only version you practiced followed this flow:

1. Ask for the first number → `prompt()`
2. Ask for the second number → `prompt()`
3. Convert input → `Number()`
4. Perform +, -, *, / calculations
5. Display results → `console.log()`

Important project note: Your interactive HTML calculator also introduced concepts you had not yet learned in the Day 1 fundamentals list, such as functions, DOM access, click events, and changing an HTML element's value. Those should be learned as the next step rather than memorized from a finished solution.

11. Quick Revision Sheet

Concept	Remember
JavaScript	Adds behaviour and logic to webpages.
let	Variable that can be reassigned.
const	Variable that cannot be reassigned.
var	Older variable declaration style.
String	Text, e.g. "Hello" or "123".
Number	Numeric value, e.g. 123 or 3.14.
Boolean	true or false.
undefined	No value has been assigned yet.
null	Intentionally empty/no value.
console.log()	Prints a value to the browser console.
prompt()	Gets user input; returned input is a String.
Number()	Converts a value to a Number.
String()	Converts a value to a String.
Boolean()	Converts a value to a Boolean.
%	Returns the remainder.
&&	AND
||	OR
!	NOT
+ with String	Can concatenate text instead of doing numeric addition.

12. Practice Questions

1. What is the difference between HTML, CSS and JavaScript?
2. What is a variable?
3. When would you choose let instead of const?
4. What is the data type of "25"? What is the data type of 25?
5. What does % do?
6. What is the difference between == and ===?
7. What does prompt() return?
8. Why might you use Number(prompt(...)) when asking for a number?
9. Predict the result of "10" + 5.

10. Predict the results of "10" - 5, "10" * 5 and "10" / 5.

> **Learning rule:** Don't try to memorize everything. Read a concept, write a tiny example, change the values, predict the output, run it, and explain why the result happened.